

Digital Image Processing

Chapter 1

1.1 What is DIP

1.3 Application of DIP (examples of Fields)

1.4 Fundamental Steps in DIP

{Pipeline from pptx}

{example: 2, edge detection, number plate detection}

1.5 Components of DIP

Read DIP1 slide full

Chapter 2 (Important)

2.3 Image Sensing

2.5 All Maths (Important)

Chapter 3 (Important)

Histogram (How & Why)

3.3 Histogram Mathematics

3.4 Spatial Filtering

Figure 3.30

Smoothing Filter, Low Pass Filter, High Pass Filter, Sharpening Filter, Avg Filter, Medium Filter

Figure 3.50

Chapter 5 (Important)

5.1 A model of the image degradation

5.2 Noise Models (Gaussian, Salt and Pepper Noise {Important among all})

5.3 Spatial Filtering (Why & How)

Noise Restoration (Why & How)

Noise Removal in Computer Vision (From [GeeksForGeeks](#))

Chapter 9

9.2 Erosion and Dilation

9.3 Opening and Closing

Chapter 10 (Important)

Image Segmentation (Why)

10.2 Filter (Laplacian Kernel Used For point detection Figure 10.4)

10.3 Thresholding (what, how)

Role of Noise in Image thresholding

10.4 Region Growing

Chapter 11

11.2 Boundary Preprocessing

Fig 11.1 with explanation

Page: 821 (Boundary Approximation using MPP algorithm)

11.5 Principal Components as Feature Descriptor

Page: 846 Texture

Table 11.3